in Figure 0.2. Both of these variants (and several others) are described and illustrated side by side in the anonymous 1458 textbook *L'Arte dell'Abbaco*, also known as the *Treviso Arithmetic*, the first *printed* mathematics book in the West.

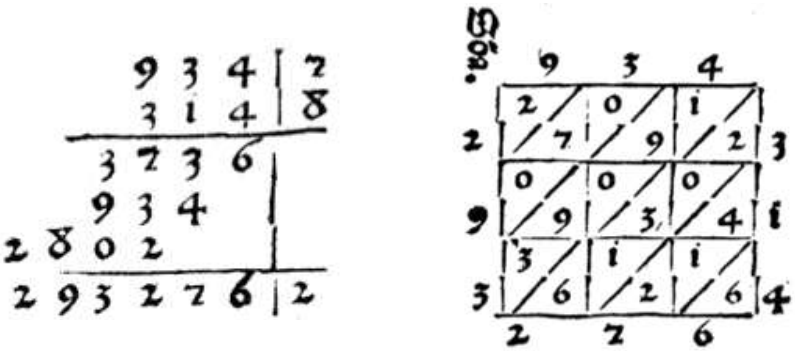

Figure 0.1. Computing $934 \times 314 = 293276$ using "long" multiplication (with error-checking by casting out nines) and "lattice" multiplication, from *L'Arte dell'Abbaco* (1458). (See Image Credits at the end of the book.)

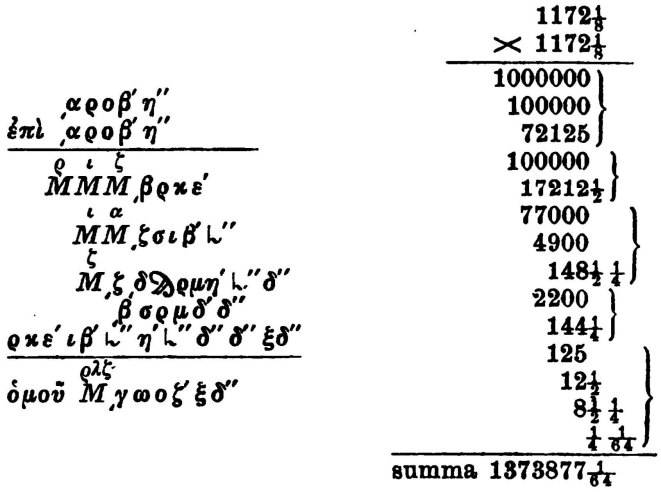

Figure 0.2. Eutocius's 6th-century calculation of $1172\frac{1}{8} \times 1172\frac{1}{8} = 1373877\frac{1}{64}$, in his commentary on Archimedes' *Measurement of the Circle*, transcribed (left) and translated into modern notation (right) by Johan Heiberg (1891). (See Image Credits at the end of the book.)

All of these variants of the lattice algorithm—and other similar variants described by Sunzi, al-Khwārizmī, Fibonacci, *L'Arte dell'Abbaco*, and many other sources—compute the product of any m-digit number and any n-digit number in $\boldsymbol{O(mn)}$ ***time***; the running time of every variant is dominated by the number of single-digit multiplications.

Duplation and Mediation

The lattice algorithm is not the oldest multiplication algorithm for which we have direct recorded evidence. An even older and arguably simpler algorithm, which does not rely on place-value notation, is sometimes called *Russian peasant multiplication*, *Ethiopian peasant multiplication*, or just ***peasant multiplication***.A

variant of this algorithm was copied into the Rhind papyrus by the Egyptian scribe Ahmes around 1650BCE, from a document he claimed was (then) about 350 years old.[10] This algorithm was still taught in elementary schools in Eastern Europe in the late 20th century; it was also commonly used by early digital computers that did not implement integer multiplication directly in hardware.

The peasant multiplication algorithm reduces the difficult task of multiplying arbitrary numbers to a sequence of four simpler operations: (1) determining parity (even or odd), (2) addition, (3) ***duplation*** (doubling a number), and (4) ***mediation*** (halving a number, rounding down).

```
PeasantMultiply(x, y):
  prod ← 0
  while x > 0
      if x is odd
          prod ← prod + y
      x ← ⌊x/2⌋
      y ← y + y
  return prod
```

x	y		$prod$
			0
123	+456	=	456
61	+912	=	1368
30	~~1824~~		
15	+3648	=	5016
7	+7296	=	12312
3	+14592	=	26904
1	+29184	=	**56088**

Figure 0.3. Multiplication by duplation and mediation

The correctness of this algorithm follows by induction from the following recursive identity, which holds for all non-negative integers x and y:

$$x \cdot y = \begin{cases} 0 & \text{if } x = 0 \\ \lfloor x/2 \rfloor \cdot (y + y) & \text{if } x \text{ is even} \\ \lfloor x/2 \rfloor \cdot (y + y) + y & \text{if } x \text{ is odd} \end{cases}$$

Arguably, this recurrence *is* the peasant multiplication algorithm. Don't let the iterative pseudocode fool you; the algorithm is fundamentally recursive!

As stated, PeasantMultiply performs $O(\log x)$ parity, addition, and mediation operations, but we can improve this bound to $O(\log \min\{x, y\})$ by swapping the two arguments when $x > y$. Assuming the numbers are represented using any reasonable place-value notation (like binary, decimal, Babylonian hexagesimal, Egyptian duodecimal, Roman numeral, Chinese counting rods, bead positions on an abacus, and so on), each operation requires at most $O(\log(xy)) = O(\log \max\{x, y\})$ single-digit operations, so the overall running time of the algorithm is $O(\log \min\{x, y\} \cdot \log \max\{x, y\}) = O(\log x \cdot \log y)$.

[10] The version of this algorithm actually used in ancient Egypt does not use mediation or parity, but it does use comparisons. To avoid halving, the algorithm pre-computes two tables by repeated doubling: one containing all the powers of 2 not exceeding x, the other containing the same powers of 2 multiplied by y. The powers of 2 that sum to x are then found by greedy subtraction, and the corresponding entries in the other table are added together to form the product.

In other words, this algorithm requires $\boldsymbol{O(mn)}$ ***time*** to multiply an m-digit number by an n-digit number; up to constant factors, this is the same running time as the lattice algorithm. This algorithm requires (a constant factor!) more paperwork to execute by hand than the lattice algorithm, but the necessary primitive operations are arguably easier for humans to perform. In fact, the two algorithms are equivalent when numbers are represented in binary.

Compass and Straightedge

Classical Greek geometers identified numbers (or more accurately, *magnitudes*) with line segments of the appropriate length, which they manipulated using two simple mechanical tools—the compass and the straightedge—versions of which had already been in common use by surveyors, architects, and other artisans for centuries. Using *only* these two tools, these scholars reduced several complex geometric constructions to the following primitive operations, starting with one or more identified reference points.

- Draw the unique line passing through two distinct identified points.
- Draw the unique circle centered at one identified point and passing through another.
- Identify the intersection point (if any) of two lines.
- Identify the intersection points (if any) of a line and a circle.
- Identify the intersection points (if any) of two circles.

In practice, Greek geometry students almost certainly drew their constructions on an *abax* (ἄβαξ), a table covered in dust or sand.[11] Centuries earlier, Egyptian surveyors carried out many of the same constructions using ropes to determine straight lines and circles on the ground.[12] However, Euclid and other Greek geometers presented compass and straightedge constructions as precise mathematical *abstractions*—points are *ideal* points; lines are *ideal* lines; and circles are *ideal* circles.

Figure 0.4 shows an algorithm, described in Euclid's *Elements* about 2500 years ago, for multiplying or dividing two magnitudes. The input consists of four distinct points A, B, C, and D, and the goal is to construct a point Z such that $|AZ| = |AC||AD|/|AB|$. In particular, if we define $|AB|$ to be our unit of length, then the algorithm computes the product of $|AC|$ and $|AD|$.

Notice that Euclid first defines a new primitive operation RightAngle by (as modern programmers would phrase it) writing a subroutine. The correctness

[11]The written numerals 1 through 9 were known in Europe at least two centuries before Fibonacci's *Liber Abaci* as "gobar numerals", from the Arabic word *ghubār* meaning dust, ultimately referring to the Indian practice of performing arithmetic on tables covered with sand. The Greek word ἄβαξ is the origin of the Latin *abacus*, which also originally referred to a sand table.

[12]Remember what "geometry" means? Democritus would later refer to these Egyptian surveyors, somewhat derisively, as *arpedonaptai* (ἀρπεδονάπται), meaning "rope-fasteners".